

Name: _____

Student ID: _____

HOMEWORK 6

Discrete Fourier Transform &

Signal Applications

2110203 COMPUTER ENGINEERING MATHEMATICS II

Generative AI Usage Declaration

- ☐ No AI Used: I did not use any AI tools to complete this homework.
- ☐ AI Assisted: I used AI tools as detailed below:

Where	AI model	How

Student Acknowledgement : I declare that I have reviewed, understood, and tested all AI-generated output. I take full responsibility for the accuracy, logic, and functionality of the final submitted work.

Signature: _____ Date: _____

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Write solutions and give explanations (1 point)

SEND ONLY PROBLEM 1. and 2.

This homework is a hybrid submission, you must submit both your completed Python notebook (.ipynb) related to the problem and this handwritten PDF (.pdf).

1. (0.6 points, 0.2 points each) Given the DFT spectrum $X[k]$, express the corresponding time-domain signal $x[n]$ in terms of its constituent real sinusoids.
 - 1.1. $X[0] = 3, X[1] = \frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}, X[2] = -2, X[3] = \frac{1}{\sqrt{2}} + j\frac{1}{\sqrt{2}}$

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1.2. $X[0] = -2, X[1] = \sqrt{3} + j1, X[2] = 3, X[3] = \sqrt{3} - j1$

1.3. $X[0] = 1, X[1] = 2 - j2\sqrt{3}, X[2] = -3, X[3] = 2 + j2\sqrt{3}$

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2. (0.4 points) Sound Denoising with FFT

https://colab.research.google.com/drive/1MVDXcgEegDXcxkac_fmIFFUpTlit6ywG?usp=sharing

In this problem, a synthesized multi-tone signal $h(t) = \sum_{i=1}^3 A_i \sin(2\pi f_i t + \phi_i)$ at a sampling rate of $f_s = 32000$ Hz is corrupted by heavy additive Gaussian and random noise. You are required to analyze the frequency spectrum using FFT and design a frequency-domain threshold filter.

2.1. [COMPLETE IN .ipynb] Plot (1) the clean signal and (2) the noisy signal with appropriate labels.

[Paste Plot of 2.1]

2.2. [COMPLETE IN .ipynb] Plot the magnitude of the FFT of the noisy signal (sort the frequency from min $\rightarrow$ max)

[Paste Plot of 2.2]

2.3. [COMPLETE IN .ipynb] Clean the noisy signal using the magnitude of the FFT.

2.4. [COMPLETE IN .ipynb] Plot the clean signal, noisy signal, and filtered signal (from your result in 2.3) with appropriate labels.

[Paste Plot of 2.4]

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- 2.5. [COMPLETE IN .ipynb] Export with IPython.display, listen to (1) the original signal, (2) the signal with noise, and (3) the signal after being filtered.
- 2.6. Write an explanation and analyze the results, Explain the mathematical filtering mechanism used. Compare the perceived audio quality and explain why FFT is effective for periodic multi-tone signals corrupted by broad-band Gaussian noise.

Answer:

Note: You will not be scored on this problem if you miss the .ipynb file.

You will not be scored too if you forget to paste the plotted image here.

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OPTIONAL, FOR PRACTICE ONLY

3. [DFT] Show that if $x[n]$ is real sequence, $X[N - k] = X^*[k]$

Problem 4-7 is the coding problem, you can practice it in colab.